ATOSIBAN

Few maternal side effects

Max 48 h

CHEMICAL NAME 1-(3-mercaptopropanoic acid)-2-(O-ethyl-D-tyrosine)-4-L-threonine-8-L-ornithine-oxytocin (acetate) — cyclic peptide

CLASS Oxytocin receptor antagonist (tocolytic)

TRADE NAMES Tractocile · Atosiban (generic) · Antocin

THERAPEUTIC USES / INDICATIONS

- Delay of imminent preterm birth (24–33 weeks, regular contractions, cervical change) — allows antenatal steroids and in-utero transfer.

ADMINISTRATION

- **IV** in three steps: bolus, high-rate infusion, then lower-rate infusion via pump.
- Total treatment max 48 h; repeat course possible if contractions recur.

DOSAGES

- **Step 1:** 6.75 mg IV bolus over 1 min.
- **Step 2:** 18 mg/h (24 mL/h of 0.75 mg/mL) for 3 h.
- **Step 3:** 6 mg/h (8 mL/h) for up to 45 h.
- Total per course preferably ≤330.75 mg; treatment ≤48 h (300 mcg/min = 18 mg/h; 100 mcg/min = 6 mg/h).

PHARMACOKINETICS

- **Onset:** contractions ↓ within 10 min. **Duration:** while infused.
- Metabolised (peptidases).
- **t½:** initial ~13 min; terminal ~1.7 h.

PHARMACODYNAMICS

- Competitive antagonist at **oxytocin (and vasopressin V1a) receptors** in myometrium → ↓ intracellular Ca^{2+} → ↓ frequency and strength of contractions.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea.
- Headache, dizziness, hot flush.
- Tachycardia, hypotension (mild).

ADVERSE EFFECTS (SERIOUS)

- Hypersensitivity, injection-site reactions.
- Postpartum haemorrhage (theoretical).
- Pulmonary oedema (rare, multiple pregnancy).

CONTRAINDICATIONS & CAUTIONS

- Gestation <24 or >33 weeks.
- Premature rupture of membranes >30 weeks.
- Intrauterine infection, antepartum haemorrhage needing delivery, eclampsia/severe pre-eclampsia, fetal death or abnormal fetal heart rate, placenta praevia/abruption.

KEY INTERACTIONS

- No significant interactions reported (used with antenatal steroids, magnesium).

NURSING MANAGEMENT

- Confirm gestation and eligibility; ensure steroids given.
- Monitor contractions, fetal heart rate (CTG), maternal BP, HR, temperature, vaginal loss.
- Prepare for transfer if needed.
- Stop if contraindication develops.

PATIENT EDUCATION

- Explains the aim: delay birth so steroids can help baby's lungs.
- Report fluid leakage, bleeding, fever or worsening pain.

KEY POINTS

- Oxytocin antagonist tocolytic.
- Well tolerated (fewer cardiac effects than β-agonists).
- Three-step IV regimen, max 48 h.
- 24–33 weeks.

SOURCES CHECKED **[672]** Atosiban 37.5 mg/5 ml concentrate for solution for infusion (UK SmPC)

Full references p. 432–457